

Gene Sequence Analysis

Lecture 7: **BLAST – Homology Search** **Pairwise Alignment**

07/06/2026

Phuc-Loi Luu, PhD

Adapted from Dr. Morgan Langille's Lecture

What is homology?

- Homology refers to shared ancestry
- Two sequences are homologous if they are derived from a common ancestral sequence
- One sequence by itself is not informative
 - it must be analyzed by comparative methods against existing sequence databases to develop hypothesis concerning relatives and function.

What is homology?

Mouse *Pax6* gene:

GTATCCAACGGTTGTGTGAGTAAAATTCTGGGCAGGTATTACGAGACTGGCTCCATCAGA

Fly *eyeless* gene:

Genetic similarity to mouse: 76.66%
Protein similarity to mouse: 100%

GTATCAAATGGATGTGTGAGCAAAATTCTCGGGAGGTATTATGAAACAGGAAGCATACGA

Shark eye control gene:

Genetic similarity to mouse: 85%
Protein similarity to mouse: 100%

GTGTCCAACGGTTGTGTCAGTAAAATCCTGGGCAGATACTATGAAACAGGATCCATCAGA

Squid eye control gene:

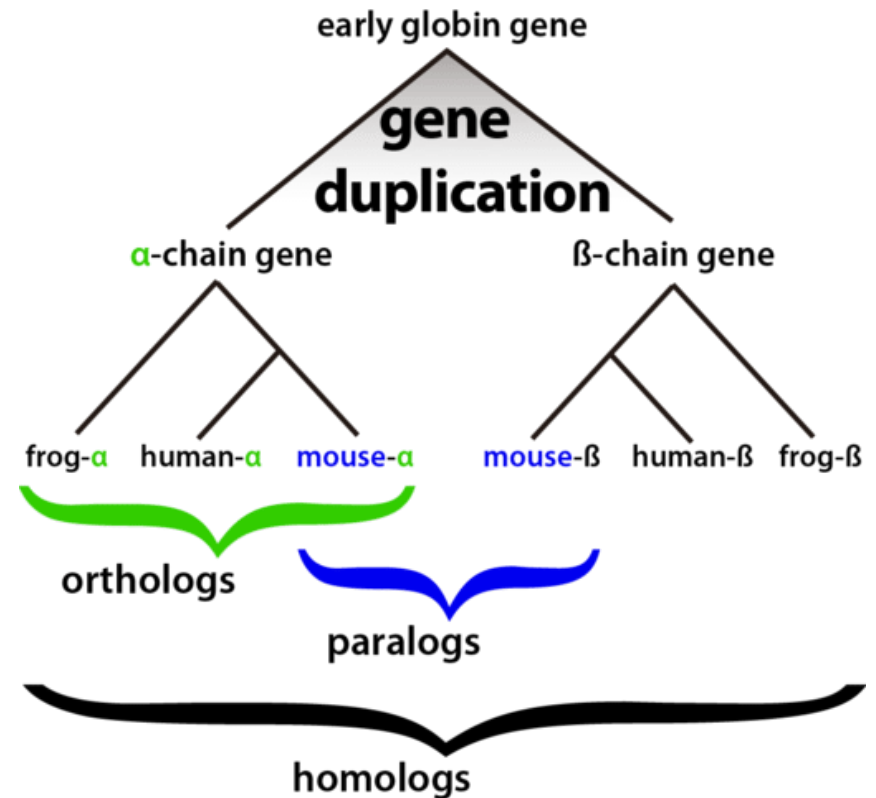
Genetic similarity to mouse: 78.33%
Protein similarity to mouse: 100%

GTCTCCAACGGCTGCGTTAGCAAGATTCTCGGACGGTACTATGAGACGGGCTCCATAAGA

Flatworm eye control gene:

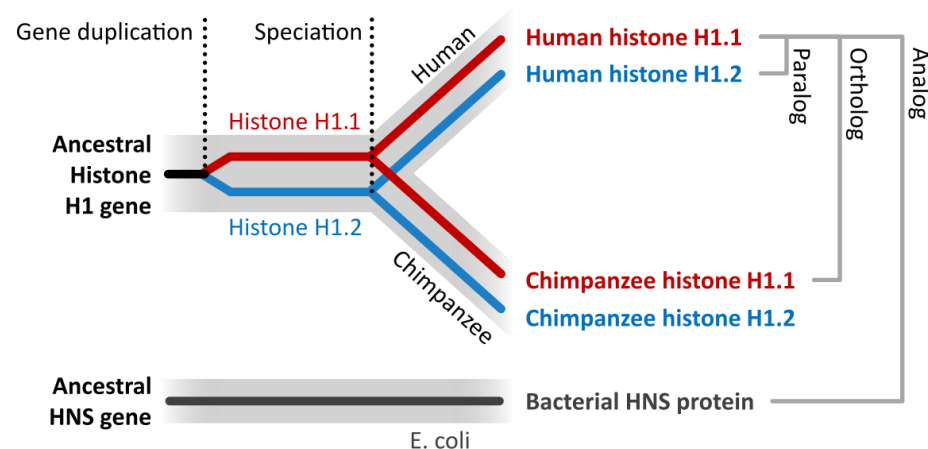
Genetic similarity to mouse: 71.66%
Protein similarity to mouse: 100%

GTGTCTAATGGTTGTGTTAGTAAAATACTTTGCCGATATTATGGAACAGGTTCTATTAAA



Types of homologs

- Orthologs
 - Think same gene in different organism
 - Often thought to have similar function
- Paralogs
 - Think gene duplication
 - Less likely to have similar function



What is similarity?

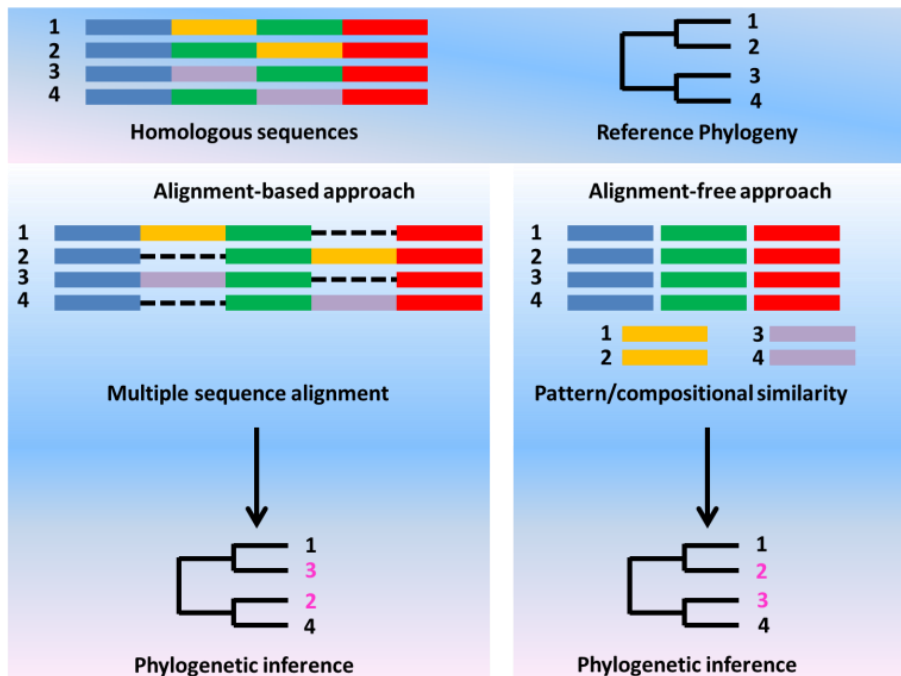
- Similarity is a measure of the likeness between sequences.
- Gene searching tools calculate the similarity between sequences and rank more similar sequences higher.
- Sequences can NOT be partially homologous
 - WRONG: Gene X is 80% homologous to Gene Y
- Sequences can be partially similar
 - CORRECT: Gene X has 80% identity to Gene Y

Identity vs Similarity

- Identity is a percentage measurement that states how many characters in the sequence are identical
- Similarity can also be used as a metric which means how many characters are “positive scoring”

Identity vs Similarity

Similarity in sequence alignment is the resemblance between two sequences when compared. This fact is dependent on the identity of sequences. Similarity depicts the extent to which the residues are aligned. Hence, similar sequences contain similar properties. In bioinformatics, similarity is a tool to assess the likeness between two proteins.



Identity in sequence alignment is the number of characters that match exactly between two different sequences. Hence, gaps do not count when assessing identity. The measurement is considered to be relational to the shorter sequence among the two sequences. It significantly implies that it has the effect where the sequence identity is not transitive. If $X=Y$ and $Y=Z$, then X is not necessarily equal to Z . This is deduced in terms of the identity distance measure.

$X = \text{AAGGCTT}$, $Y = \text{AAGGC}$ and $Z = \text{AAGGCAT}$.

Identity between X and Y is 100% {5 identical nucleotides / $\min[\text{length}(X), \text{length}(Y)]$ }.

Identity between Y and Z is also 100%.

But identity between X and Z is only 85% {(6 identical nucleotides / 7)}.

Selected annotation from match [P03372](#)

Accession	Entry name	Status	Protein names	Organism	Length	Gene names
P03372	ESR1_HUMAN	★	Estrogen receptor	Homo sapiens (Human)	595	ESR1 ESR NR3A1

Alignment 1 against [P03372](#)

Score	218	E-value	4.0×10^{-21}
Identity	48.0%	Positives	68.0%
Query length	76	Match length	595
Position	P03372 matches from 183 to 252 (70AA), in the query sequence from 2 to 71 (70AA)		

2	RICGVCGRATGFHFNAMTCEGCKGFFRRSMKRKALFTCPFNGDCRITKDNRRHCQACRL	61 Query
	R C VC D A+G+H+ +CEGCK FF+RS++ + CP C I K+ R+ CQACRL	
183	RYCAVCNDYASGYHYGVNSCEGCKAFFKRSIQGHNDYMCPTNQCTIDKNRRKSCQACRL	242 P03372
62	KRCVDIGMMK	71 Query
	++C ++GMMK	
243	RKCYEVGMMK	252 P03372

Identity vs Similarity

Identity

The extent to which two (nucleotide or amino acid) sequences are **invariant** (identical).

Similarity

The extent to which (nucleotide or amino acid) sequences are **related**. The extent of similarity between two sequences can be based on percent sequence identity and/or conservation. In BLAST similarity refers to a positive matrix score. This is quite flexible (see later examples of DNA polymerases) – similar across the whole sequence *or* similarity restricted to domains !

Homology

Similarity attributed to **descent from a common ancestor**.

Assessing Sequence Similarity

Rbn KETAAAKFERQHMD

Lsz KVFGRCELAAAMKRHGLDNYRGYSLGNWVCAAKFESNFNT

Rbn SST SAASSSNYCMMKSRNLTKDRCKEMNTFVHESLA
Lsz QATNRNTDGSTDYGILQINSRWWCNDGRTP GSRN

Rbn DVQAVCSQKNVACKNGQTNCYQSYSTMSTIDCRETGSSKY
Lsz LCNIPCSALLSSDITASVNC AKKIVSDGDGMNAWVAWR

Rbn PNACYKTTQANKHITVACEGNPYVPHF¹DASV
Lsz NRCKGTDVQA WIRG¹CR¹L

is this alignment significant?

DNA scoring systems

Sequence 1 ACTACCAGTTCATTTGATACTTCTCAAA
Sequence 2 TACCATTACCGTGTTAACTGAAAGGACTTAAAGACT

 | | | | |

	A	C	G	T
A	1	0	0	0
C	0	1	0	0
G	0	0	1	0
T	0	0	0	1

Match: 5 x 1 = 5

Mismatch: 19 x 0 = 0

Score: 5

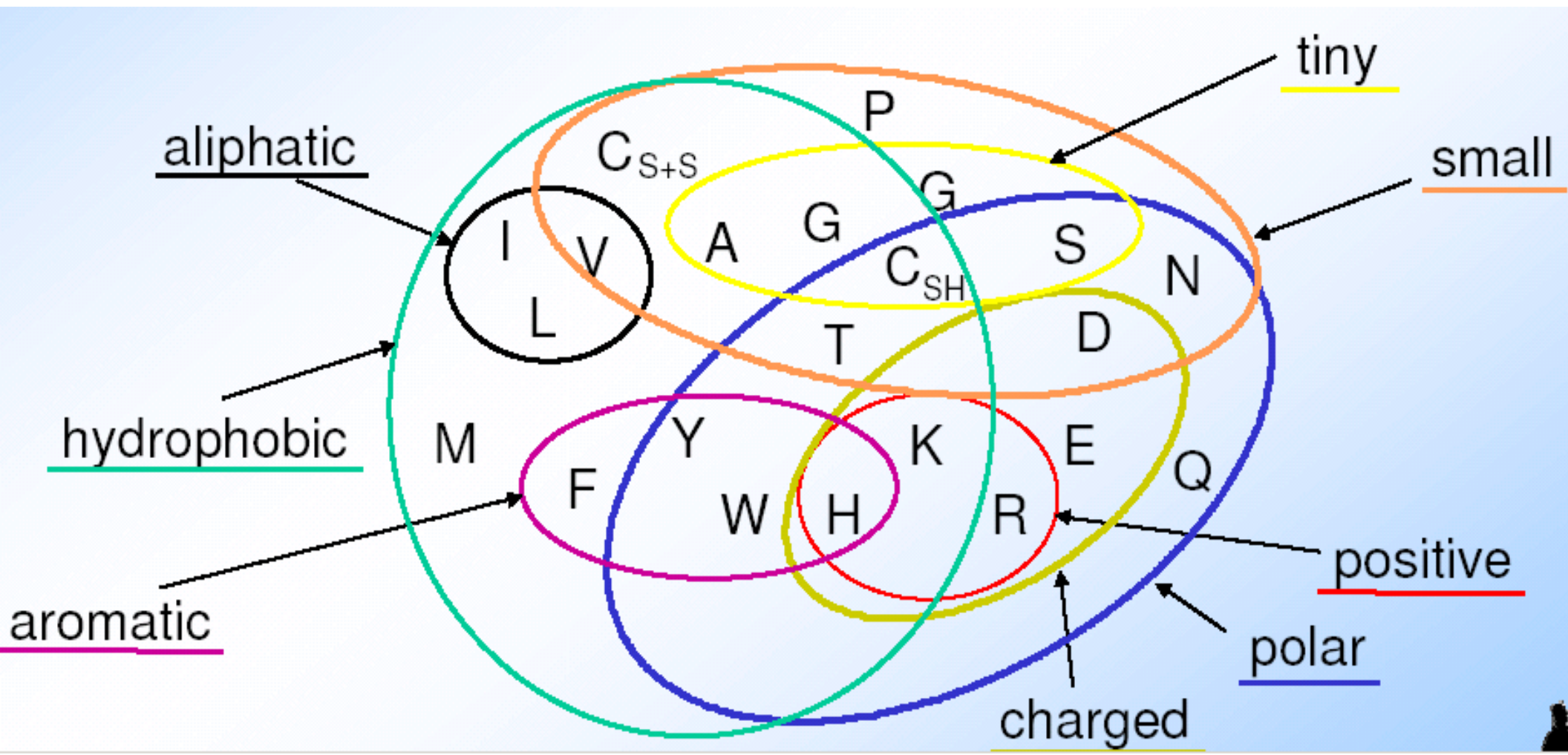
The Scoring Schemes or Weight Matrices

Genetic Code Scoring

- Fitch 1966 based on Nucleotide Base change required (0,1,2,3)
- Required to interconvert the codons for the two amino acids
- Rarely used nowadays

Complication: „inexact“ is not binary (1|0) but something *relative*

Amino acids have different physical and biochemical properties that are/are not important for function and thus influence their probability to be replaced in evolution



The Scoring Schemes or Weight Matrices

Chemical Similarity Scoring

- ❖ Similarity based on Physio-chemical properties
- ❖ MacLachlan 1972, Based on size, shape, charge and polar
- ❖ Score 0 for opposite (e.g. E & F) and 6 for identical character

The Scoring Schemes or Weight Matrices

Observed Substitutions or PAM matrices

- ❖ Based on Observed Substitutions
- ❖ Chicken and Egg problem
- ❖ Dayhoff group in 1977 align sequence manually
- ❖ Observed Substitutions or point mutation frequency
- ❖ MATRICES are PAM30, PAM250, PAM100 etc

A**I****L****D****C****T****G****R****T****G**.....

A**L****L****D****C****T****G****R**--.....

S**L****I****D****C****S****A****R**-G.....

A**I****L****N****C****T****L**-R**G**.....

Where does the score (S) come from?

- The quality of each pair-wise alignment is represented as a score and the scores are ranked.
- **Scoring matrices** are used to calculate the score of the alignment base by base (DNA) or amino acid by amino acid (protein).
- **The alignment score will be the sum of the scores for each position.**

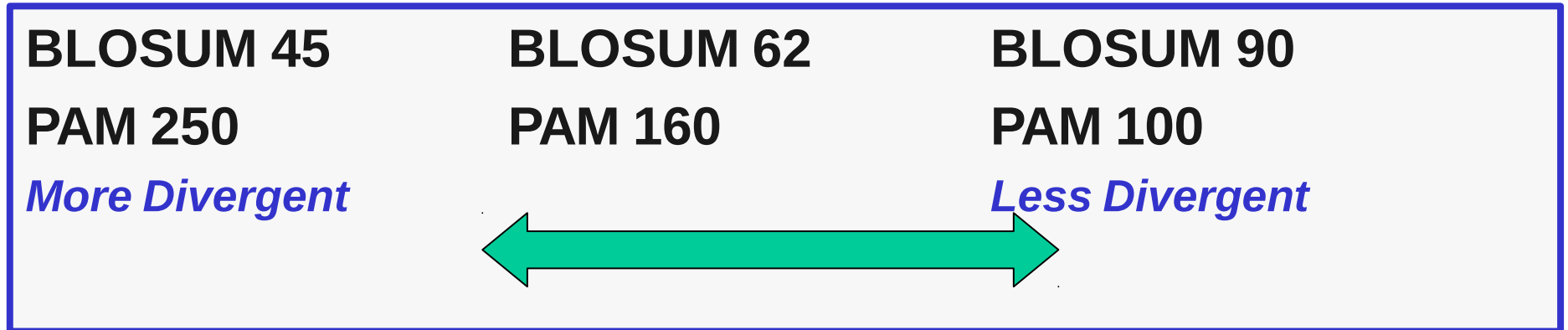
What's a scoring matrix?

- Substitution matrices are used for amino acid alignments.
 - each possible residue substitution is given a score
- A simpler unitary matrix is used for DNA pairs
 - each position can be given a score of +1 if it matches and a score of -1 if it does not.

	A	C	D	E	F	G	H
A	4	0	-2	-1	-2	0	-2
C	0	9	-3	-4	-2	-3	-3
D	-2	-3	6	2	-3	-1	-1
E	-1	-4	2	5	-3	-2	0
F	-2	-2	-3	-3	6	-3	-1
G	0	-3	-1	-2	-3	7	-2
H	-2	-3	-1	0	-1	-2	8

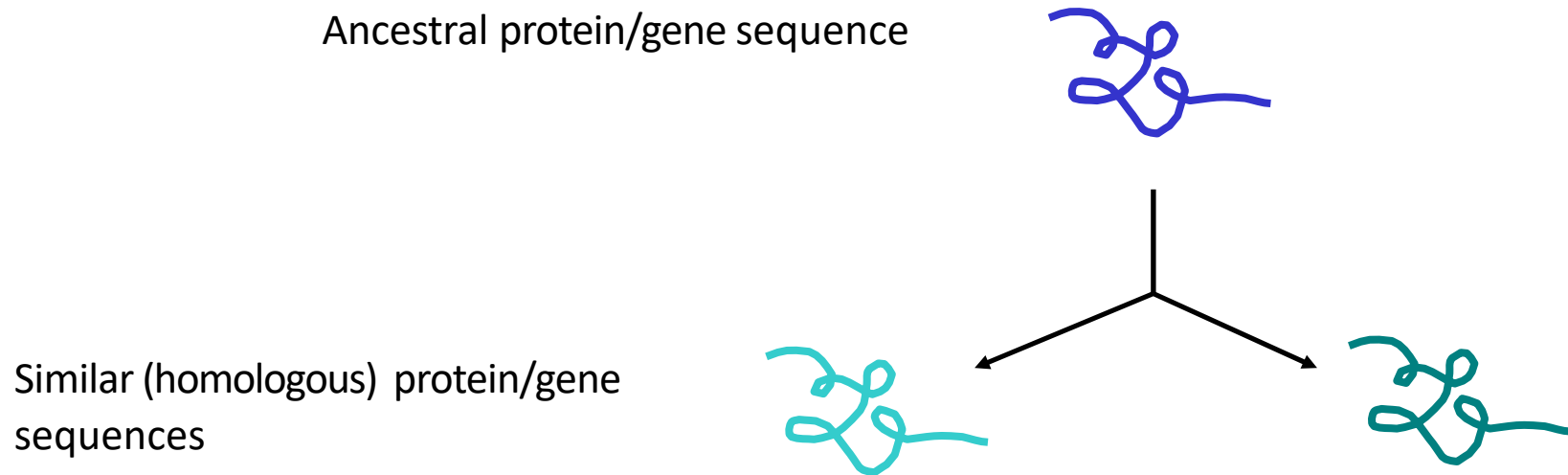
BLOSUM 62

BLOSUM vs. PAM



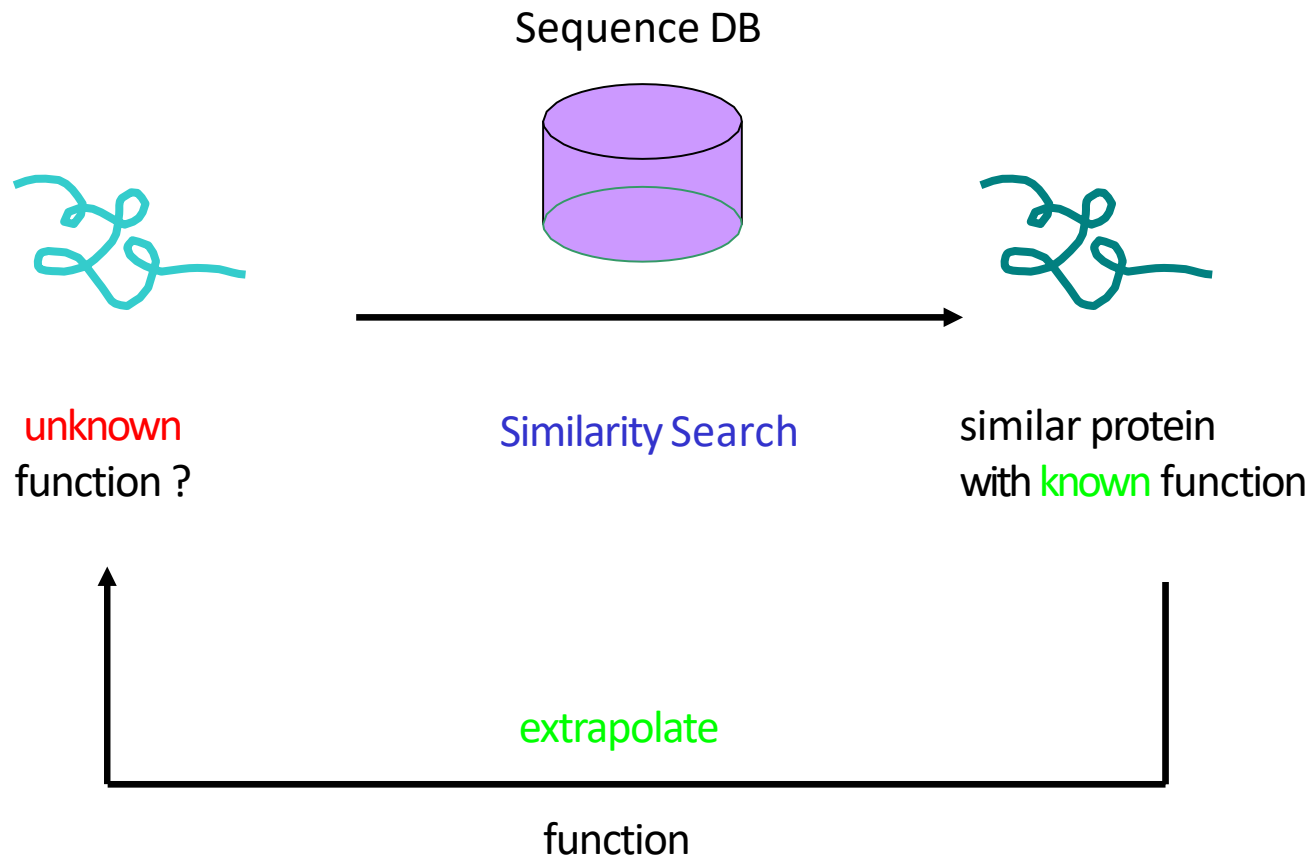
- BLOSUM62 is the default matrix in BLAST. Though it is tailored for comparisons of moderately distant proteins, it performs well in detecting closer relationships. A search for distant relatives may be more sensitive with a different matrix.

Importance of Similarity



Similar sequences: probably have the same ancestor, share the same structure, and have a similar biological function

Importance of Similarity



QUERY sequence(s)

```
>gi|15237380|ref|NP_197163.1| myb family transcription factor (MYB43) [Arabidopsis thaliana]  
MGRQPCCDKVGLKKGPWTIEEDKKLINFILTNHGHCWRALPKLSGLRCGKSCRLRWINYLRPDLKRGLL  
SEYEEQKVINLHAQLGNRWSKIASHLPGRTDNEIKNHWNTHIKKKLRKMGIDPLTHKPLSEQEASQQAQG  
RKKSIVPHDDKNPKQDQQTDEQEQLLEALEKNNTSVSGDGFIDEVPLLPHEILIDISSHHHHSN  
DDNVNINTSKFTSPSSSSSTSSCISVVPGDEFKFFDEMEILDKLWLSDDSLGDDISKDGKFNNTV  
DTMNLWDINDLSSLDMMNEHDDGFIGNGNGCSRMLDQDSWTFDLL
```

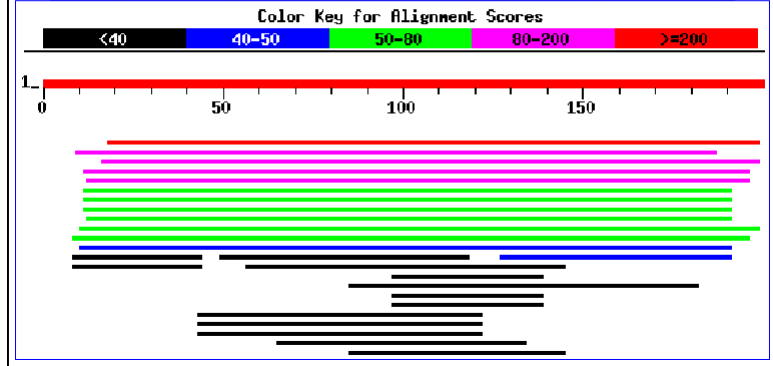
BLAST program

BLAST
database

BLAST results

Distribution of 26 Blast Hits on the Query Sequence

P75430 Hypothetical protein MG245 homolog (H91_orf164)..S=46.2 E=5e-05



Considerations for choosing a BLAST database

- First consider your research question:
 - Are you looking for an particular gene in a particular species?
 - BLAST against the genome of that species.
 - Are you looking for additional members of a protein family across all species?
 - BLAST against the non-redundant database (nr), if you can't find hits check wgs, htgs, and the trace archives.
 - Are you looking to annotate genes in your species of interest?
 - BLAST against known genes (RefSeq) and/or ESTs from a closely related species.

When choosing a database for BLAST...

- Changing your choice of database is changing your search space
- Database size affects the BLAST statistics
- Databases change rapidly and are updated frequently

Importance of Similarity

Rule-of-thumb:

If your sequences are more than **100 amino acids** long (or 100 nucleotides long) you can consider them as homologues if **25%** of the **aa** are identical (**70%** of **nucleotide** for DNA). Below this value you enter the **twilight zone**.

Twilight zone = protein sequence similarity between ~0-20% identity: is **not** statistically **significant**, i.e. could have arisen by chance.



Beware:

- E-value (*Expectation value*)
- length of the segments similar between the two sequences
- The number of insertions/deletions

BLAST, FASTA, SSEARCH, and other commonly used similarity searching programs produce accurate statistical estimates that can be used to reliably infer homology. Searches with protein sequences (BLASTP, FASTP, SSEARCH,) or translated DNA sequences (BLASTX, FASTX) are preferred because they are 5- to 10-fold more sensitive than DNA:DNA sequence comparison. The 30% identity rule-of-thumb is too conservative; statistically significant [$E() < 10^{-6} - 10^{-3}$] protein homologs can share less than 20% identity. E()-values and bit scores (bits >50) are far more sensitive and reliable than percent identity for inferring homology.

Descriptions

Sequences producing significant alignments:

Select: [All](#) [None](#) Selected:0

[Alignments](#) [Download](#) [GenBank](#) [Graphics](#) [Distance tree of results](#)

Description	Max score	Total score	Query cover	E value	Ident	Accession
☐ Homo sapiens laminin subunit alpha 4 (LAMA4), transcript variant 2, mRNA	11596	11596	100%	0.0	99%	NM_002290.4
☐ Homo sapiens laminin, alpha 4, mRNA (cDNA clone MGC:74960 IMAGE:6164247), complete cds	11533	11533	100%	0.0	99%	BC066552.1
☐ Homo sapiens laminin subunit alpha 4 (LAMA4), transcript variant 3, mRNA	11516	11516	100%	0.0	99%	NM_001105207.2
☐ Homo sapiens laminin subunit alpha 4 (LAMA4), transcript variant 1, mRNA	11498	11498	100%	0.0	99%	NM_001105206.2
☐ laminin alpha 4 chain [human, fetal lung, mRNA, 6204 nt]	11407	11407	98%	0.0	99%	S78569.1
☐ PREDICTED: Gorilla gorilla gorilla laminin, alpha 4, transcript variant 1 (LAMA4), mRNA	11383	11383	100%	0.0	99%	XM_004044552.1
☐ PREDICTED: Homo sapiens laminin, alpha 4 (LAMA4), transcript variant X1, mRNA	11367	11367	99%	0.0	99%	XM_005266983.3
☐ PREDICTED: Gorilla gorilla gorilla laminin, alpha 4, transcript variant 2 (LAMA4), mRNA	11311	11311	100%	0.0	99%	XM_004044553.1
☐ PREDICTED: Gorilla gorilla gorilla laminin, alpha 4, transcript variant 3 (LAMA4), mRNA	11287	11287	100%	0.0	99%	XM_004044554.1
☐ PREDICTED: Homo sapiens laminin, alpha 4 (LAMA4), transcript variant X2, mRNA	11236	11236	97%	0.0	99%	XM_005266984.3
☐ H.sapiens mRNA for laminin alpha 4 protein	11158	11158	95%	0.0	100%	X91171.1
☐ Homo sapiens mRNA for LAMA4 variant protein, clone: hh01833	10966	10966	95%	0.0	99%	AB210027.1
☐ PREDICTED: Pan paniscus laminin, alpha 4 (LAMA4), transcript variant X2, mRNA	10924	10924	96%	0.0	99%	XM_003822216.3
☐ PREDICTED: Pan troglodytes laminin, alpha 4 (LAMA4), transcript variant X2, mRNA	10890	10890	96%	0.0	99%	XM_518696.5
☐ PREDICTED: Pan paniscus laminin, alpha 4 (LAMA4), transcript variant X1, mRNA	10852	10852	96%	0.0	99%	XM_008975760.1
☐ PREDICTED: Pan paniscus laminin, alpha 4 (LAMA4), transcript variant X3, mRNA	10831	10831	96%	0.0	99%	XM_008975761.1
☐ PREDICTED: Pan troglodytes laminin, alpha 4 (LAMA4), transcript variant X1, mRNA	10816	10816	96%	0.0	99%	XM_009451861.1

[Download](#) [GenBank](#) [Graphics](#)

Homo sapiens laminin subunit alpha 4 (LAMA4), transcript variant 2, mRNA

Sequence ID: [ref|NM_002290.4|](#) Length: 7355 Number of Matches: 1

Range 1: 116 to 6412 [GenBank](#) [Graphics](#)

[Next Match](#) [Previous Match](#)

Score	Expect	Identities	Gaps	Strand
11596 bits(6279)	0.0	6291/6297(99%)	0/6297(0%)	Plus/Plus

Query 1 AGAAGGTAAAAAGGGAGTGGTGAGAATGAATGTGAGAAGGAAGCCAGGACAGCGCAGTCC 60
 Sbjct 116 AGAAGGTAAAAAGGGAGTGGTGAGAATGAATGTGAGAAGGAAGCCAGGACAGCGCAGTCC 175

Query 61 CCAGTCCCCGAACGGCCAGGGAGAGGAGGTGGCCTAGCGCTGGCGGGGCTACCCCCAATCC 120
 Sbjct 176 CCAGTCCCCGAACGGCCAGGGAGAGGAGGTGGCCTAGCGCTGGCGGGGCTACCCCCAATCC 235

Query 121 GTCTGCCTTTTGATGCCGTACTCTGCTGGTTGCGCAGCCACCTCGGGATACTGCACACGG 180
 Sbjct 236 GTCTGCCTTTTGATGCCGTACTCTGCTGGTTGCGCAGCCACCTCGGGATACTGCACACGG 295

Query 181 AGAGGAGGGAAAATAAGCGAGGCACCGCCGCACCACGCGGGAGACCTACGGAGACCCACA 240
 Sbjct 296 AGAGGAGGGAAAATAAGCGAGGCACCGCCGCACCACGCGGGAGACCTACGGAGACCCACA 355

Query 241 GCGCCCCGAGCCCTGGAAGAGCACTACTGGATGTCAGCGGAGAAATGGCTTTGAGCTCAGC 300
 Sbjct 356 GCGCCCCGAGCCCTGGAAGAGCACTACTGGATGTCAGCGGAGAAATGGCTTTGAGCTCAGC 415

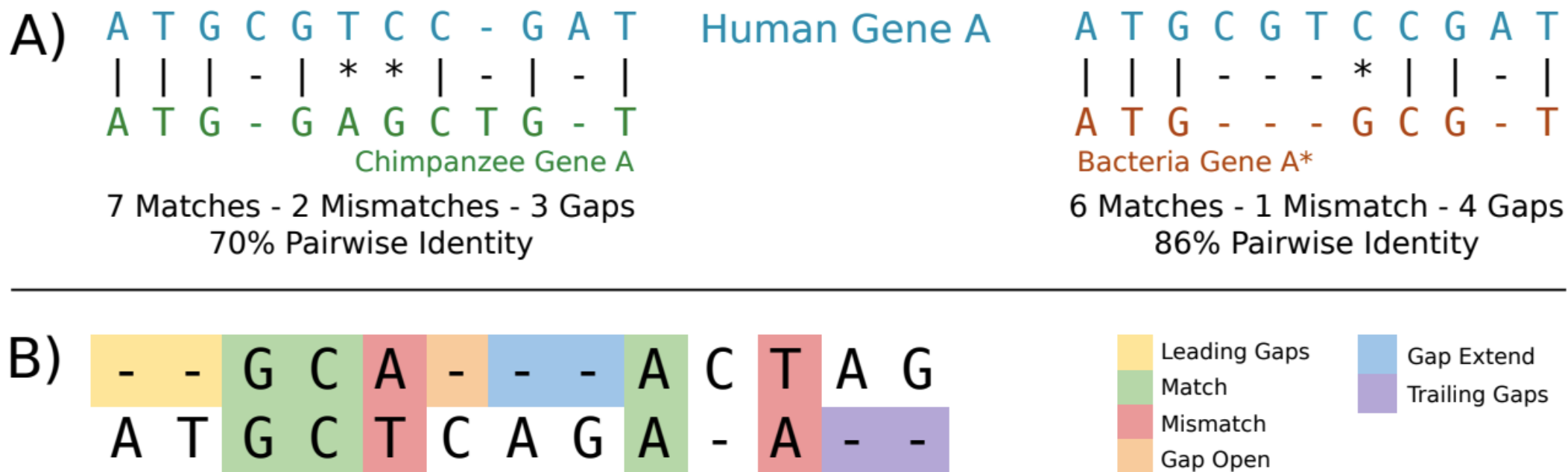


FIGURE 1.2: Similarity versus homology and alignment scoring: *A)* Two sequence alignments to Human Gene A highlighting how true homologs (Chimpanzee Gene A) can have similar similarity to non-homologous sequences (Bacteria Gene A*). *B)* A sequence alignment with different reward or penalties highlighted to illustrate how homology search algorithms score alignments based on sequence similarity.

Doolittle's twilight zone is between 15-25% and is a 'rule of thumb' for sequences of 100 amino acid length or greater (29). The 100 amino acids caveat is due to the greater probability of short sequences having the same amino acid sequence by chance and therefore the level of confidence in short alignments is not as easy to give general advice for (19, 29). No quantitative experiment was done to determine this

As a part of a study of structural ncRNAs that were used to benchmark multiple sequence alignment programs the twilight zone for nucleotide alignment accuracy was calculated (11). It was between 50-60% which is much higher than the 10-20% calculated by Thompson *et al.* 1999 for protein alignments (32). This zone represents

billion years ago (e.g. humans to bacteria). Moreover, DNA:DNA alignment statistics are less accurate than protein:protein statistics; while protein:protein alignments with expectation values < 0.001 can reliably be used to infer homology, DNA:DNA expectation values $< 10^{-6}$ often occur by chance, and 10^{-10} is a more widely accepted threshold for homology based on DNA:DNA searches. The most effective way to improve search sensitivity with DNA sequences is to use translated-DNA:protein alignments, such as those produced by BLASTX and FASTX, rather than DNA:DNA alignments.

Sequence Similarity Searching – The statistics are important

- Discriminating between real and artifactual matches is done using an estimate of probability that the match might occur by chance.

What do the Score and the e-value really mean?

- The **quality** of the alignment is represented by the Score.
 - **Score (S)**
 - The score of an alignment is calculated as the sum of substitution and gap scores. Substitution scores are given by a look-up table (PAM, BLOSUM) whereas gap scores are assigned empirically .
- The **significance** of each alignment is computed as an E value.
 - **E value (E)**
 - Expectation value. The number of different alignments with scores equivalent to or better than S that are expected to occur in a database search by chance. The lower the E value, the more significant the score.

I'm confused! What does the E-value mean again?

- **E value (E)**
 - Expectation value. The number of different alignments with scores equivalent to or better than S that are expected to occur in a database search by chance. The lower the E value, the more significant the score.
- When $E < 0.01$, P -values and E -value are nearly identical.
 - So, the E-value is the number of times you expect to see your hit occur in the database (with as good as or better score) due to random chance alone.

Notes on E-values

- Low E-values suggest that sequences are homologous
 - Can't show non-homology
- Statistical significance depends on both the size of the alignments and the size of the sequence database
 - Important consideration for comparing results across different searches
 - E-value increases as database gets bigger
 - E-value decreases as alignments get longer